

MARTINA ALUTTO

Postdoctoral Researcher
at KTH Royal Institute of Technology

alutto@kth.se | martinalutto.github.io | Malvinas väg 10, 114 28, Stockholm, Sweden

RESEARCH EXPERIENCE

Postdoctoral Researcher **2025 – current**
KTH Royal Institute of Technology, Stockholm, Sweden
Supervisors: ass. prof. Angela Fontan and prof. Karl Henrik Johansson
Research topics: Social networks, opinion dynamics, network systems
Affiliated with the Wallenberg AI, Autonomous Systems and Software Program (WASP).

Research Fellow **2024 – 2025**
National Research Council (CNR), Torino, Italy
Supervisors: prof. Chiara Ravazzi and prof. Fabrizio Dabbene
Analysis and control of innovation adoption models coupled with opinion dynamics.

PhD Student in Pure and Applied Mathematics **2021 – 2024**
Politecnico di Torino – Università di Torino, Torino, Italy
Supervisors: prof. Fabio Fagnani and prof. Giacomo Como
Thesis: *Network Effects and Behavioral Feedback in Epidemic Models*

RESEARCH VISITS

University of Tokyo, Tokyo, Japan **06/2025 – 07/2025**
Supervisor: prof. Hideaki Ishii
Control of innovation adoption-opinion models (ASPIRE program).

Kyushu University, Fukuoka, Japan **07/2025**
Supervisor: prof. Yoshio Ebihara

Cornell University, Ithaca (NY), USA **09/2023 – 02/2024**
Supervisor: prof. Francesca Parise
Optimal control problems on epidemic models.

EDUCATION

Ph.D. in Pure and Applied Mathematics **2021 – 2025**
Politecnico di Torino – Università di Torino, Torino, Italy
Final evaluation: cum laude
Thesis: *"Network Effects and Behavioral Feedback in Epidemic Models"*
Supervisors: prof. Fabio Fagnani and prof. Giacomo Como

M.Sc. in Mathematical Engineering **2018 – 2021**
Politecnico di Torino, Torino, Italy — Final grade: 110/110 cum laude
Thesis: *"Feedback Control Policies and Network Effects in Epidemics Models"*
Thesis supervisors: prof. Fabio Fagnani and prof. Giacomo Como

B.Sc. in Mathematics for Engineering **2015 – 2018**
Politecnico di Torino, Torino, Italy

PUBLICATIONS

Journal Papers

1. **M. Alutto**, S. Bellotti, F. Dabbene and C. Ravazzi, "Modeling and Control of Sustainable Transitions through Opinion-Behavior Coupling in Heterogeneous Networks," *IEEE Transactions on Computational Social Systems*, 2026, doi: 10.1109/TCSS.2026.3715278.
2. **M. Alutto**, L. Cianfanelli, G. Como and F. Fagnani, "Network Behavioral-Feedback SIR Epidemic Model," *Automatica*, vol. 193, Art. 113220, 2026, doi: 10.1016/j.automatica.2026.113220.
3. **M. Alutto**, L. Cianfanelli, G. Como and F. Fagnani, "On the Dynamic Behavior of the Network SIR Epidemic Model," *IEEE Transactions on Control of Network Systems*, vol. 12, no. 1, pp. 177–189, 2025, doi:

10.1109/TCNS.2024.3448136.

Preprints

1. **M. Alutto**, F. Dabbene, A. Fontan, K. H. Johansson and C. Ravazzi, “Competition, Switching, and Control in Coupled Adoption-Opinion Dynamics,” preprint, 2026.
2. **M. Alutto**, L. Cianfanelli, G. Como, F. Fagnani and F. Parise, “Optimal Control for Behavioral-Feedback SIR Epidemic Model,” preprint arXiv:2512.08544, 2026.
3. **M. Alutto**, Q. Xu, F. Dabbene, H. Ishii and C. Ravazzi, “Predictive Control Strategies for Sustaining Innovation Adoption on Multilayer Social Networks,” preprint arXiv:2509.01457, 2025.

Conference Papers

1. **M. Alutto**, L. Zino, K. H. Johansson and A. Fontan, “Who Leads the Crowd? An Experimental Proof-of-Concept for Modeling Emergent Leadership in Opinion Dynamics,” accepted for presentation and publication in the Proc. of the IFAC Workshop on Cyber-Physical & Human Systems, 2026.
2. **M. Alutto**, L. Zino, K. H. Johansson and A. Fontan, “On Leadership Emergence in Opinion Dynamics on Social Networks,” accepted for presentation and publication in the Proc. of the 65th IEEE Conference on Decision and Control (CDC 2026), preprint arXiv:2606.05825, 2026.
3. **M. Alutto**, L. Zino, K. H. Johansson and A. Fontan, “On a Co-evolving Opinion-Leadership Model in Social Networks,” Proc. of the 23rd IFAC World Congress, Busan, Republic of Korea, 2026.
4. **M. Alutto**, C. Ravazzi, F. Dabbene, K. H. Johansson and A. Fontan, “On a Coupled Adoption-Opinion Framework for Competing Innovations,” Proc. of the 23rd IFAC World Congress, Busan, Republic of Korea, 2026.
5. **M. Alutto**, Q. Xu, F. Dabbene, H. Ishii and C. Ravazzi, “Model Predictive Control for Coupled Adoption-Opinion Dynamics,” Proc. of the 64th IEEE Conference on Decision and Control (CDC 2025), Rio de Janeiro, Brazil, 2025.
6. **M. Alutto**, L. Cianfanelli, G. Como, F. Fagnani and F. Parise, “Behavioral-Feedback SIR Epidemic Model: Analysis and Control,” Proc. of the 64th IEEE Conference on Decision and Control (CDC 2025), Rio de Janeiro, Brazil, 2025.
7. **M. Alutto**, L. Cianfanelli, G. Como and F. Fagnani, “Multiple Peaks in Network SIR Models,” Proc. of the 61st IEEE Conference on Decision and Control (CDC 2022), Cancún, Mexico, 2022.
8. **M. Alutto**, G. Como and F. Fagnani, “On SIR Epidemic Models with Feedback-Controlled Interactions and Network Effects,” Proc. of the 60th IEEE Conference on Decision and Control (CDC 2021), online, 2021.

PhD Thesis

1. **M. Alutto**, “Network Effects and Behavioral Feedback in Epidemic Models,” PhD thesis, Politecnico di Torino – Università di Torino, 2025.

CONFERENCE PRESENTATIONS & SEMINARS

Conference presentations

- *On a Co-Evolving Opinion-Leadership Model in Social Networks* and *On a Coupled Adoption-Opinion Framework for Competing Innovations*, 23rd IFAC World Congress, Busan, Republic of Korea, August 2026.
- *Network Effects and Behavioral Feedback in Epidemic Models*, 14th European Conference on Mathematical & Theoretical Biology (ECMTB 2026), University of Graz, Austria, July 2026 (invited minisymposium talk).
- *Behavioral-Feedback SIR Epidemic Model: Analysis and Control* and *Model Predictive Control for Coupled Adoption-Opinion Dynamics*, 64th IEEE Conference on Decision and Control (CDC 2025), Rio de Janeiro, Brazil, December 2025.
- *Network Effects and Behavioral Feedback in Epidemics*, SIAM Conference on Applications of Dynamical Systems (DS25), Denver, USA, May 2025.
- *Multiple Peaks in Network SIR Epidemic Models*, 61st IEEE Conference on Decision and Control (CDC 2022), Cancún, Mexico, December 2022.
- *Multi-modal Behaviours in Network SIR Model*, 25th International Symposium on Mathematical Theory of Networks and Systems (MTNS 2022), Bayreuth, Germany, September 2022.
- *On SIR Epidemic Models with Feedback-Controlled Interactions and Network Effects*, 60th IEEE Conference on Decision and Control (CDC 2021), online, December 2021.

Posters

- *On Leadership Emergence in Opinion Dynamics on Social Networks*, 3rd Digital Futures Young Scientist Conference, Stockholm, Sweden, March 2026.

Other invited talks and seminars

- Invited seminars during the ASPIRE research visit at the University of Tokyo, Kyushu University, and the RACOT Seminar, Japan, July 2025.
- Seminar hosted by prof. Karl Henrik Johansson's group, Department of Decision and Control Systems, KTH, October 2024.
- Invited lecture in "Topics in Dynamical Systems" (MATH 7280), prof. Steven Strogatz, Cornell University, November 2023.
- FIND Graduate Day, School of Electrical and Computer Engineering, Cornell University, September 2023.

AWARDS & MEMBERSHIPS

- Young Talent Project Scholarship, Politecnico di Torino and Fondazione CRT, 2014.
- IFAC – Affiliate since 2026.
- IEEE Technical Committee on Network Systems – Member since 2025.
- GNAMPA (Gruppo Nazionale per l'Analisi Matematica, la Probabilità e le loro Applicazioni), Istituto Nazionale di Alta Matematica – Member since 2024.
- IEEE Control Systems Society (CSS) – Member since 2023.
- IEEE – Member since 2023.
- IEEE Young Professionals – Member since 2023.

OTHER ACTIVITIES

- Supervision of master's theses and research internships:
  - David Puustinen (KTH, 2025–present), *"Analyzing Opinion Spreading in Dynamic Signed Social Networks Using Real-World Data,"* co-supervised with Dr. Pelin Şekercioğlu.
  - Jakob Götberg (KTH, 2026), *"Competing Technology Adoption in Networked Systems using Real-World Data,"* research internship.
  - Sofia Bellotti (Politecnico di Torino, 2025), *"Estimating Innovation Adoption Parameters in a Coupled Opinion–Adoption Model,"* co-supervised with prof. Chiara Ravazzi and prof. Fabrizio Dabbene.
  - Davide Sipione (Politecnico di Torino, 2024), *"On Convexity and Stability of Multicommodity Dynamical Flow Networks,"* co-supervised with prof. Giacomo Como.
- Reviewer for international journals and conference proceedings:
  - Journals (29): Automatica, IEEE Transactions on Automatic Control, IEEE Transactions on Control of Network Systems, Nonlinear Analysis: Hybrid Systems, European Journal of Control, International Journal of Control, Applied Probability Journals, Review Commons, IEEE Open Journal of Control Systems, IEEE Control Systems Letters, PLOS Computational Biology, SIAM Journal of Life Sciences.
  - Conference proceedings (12): IEEE Conference on Decision and Control, IEEE American Control Conference, IEEE European Control Conference, IFAC World Congress.
- Organizer of a proposed invited session "Analysis and Control of Complex Socio-Technical Systems in the Social and Life Sciences" at the 25th European Control Conference (ECC), Brussels, Belgium, 2027.
- Member of the organizing team of the workshop "Algorithmic Game Theory, Mechanism Design, and Learning," Politecnico di Torino, November 2022.

SKILLS

Languages	**Italian** (native); **English** (fluent)
Programming	Python, LaTeX, C, C++, R, Matlab